Ministry for Primary Industries
Manatū Ahu Matua

STATEMENT OF WORK

for a research project:

407609 INV2026-01: Testing the viability of Baited Remote Underwater Video (BRUV) to monitor trends in marine finfish and biodiversity in the Fiordland Marine Area

under the

PANEL AGREEMENT FOR INSHORE & HIGHLY MIGRATORY FISHERIES RESEARCH

Between

MINISTRY FOR PRIMARY INDUSTRIES

and

Sea Through Science

PARTIES

HIS MAJESTY THE KING in right of New Zealand acting by and through Dan Bolger, Deputy Director General, Fisheries New Zealand, Ministry for Primary Industries **("MPI")**

AND

Sea Through Science Limited (NZBN9429051718091) and having its registered offices at 39 Amorino Drive, Red Beach 0932 (the "Contractor").

BACKGROUND

A. MPI and the Contractor are parties to a Panel Services Panel Agreement entitled "[Highly Migratory and Inshore Fisheries Research, Agreement Number C0038158]" dated 18/08/2025 (**"Panel Agreement"**) under which they may from time to time enter into SOWs.

B. This document is a SOW for the purposes of the Panel Agreement. Terms used in this SOW have the meanings given to them in the Panel Agreement. The Panel Agreement applies to and governs this SOW as if set out in full herein in its entirety. References in this SOW to Schedule 1 and Schedule 2 are to schedules of the Panel Agreement.

C. MPI requires a panel of research providers to undertake fisheries research relating to inshore and highly migratory fish species. Under this SOW the Contractor will provide research services in the following categories as follows:

- Any other services reasonably related to highly migratory and inshore fisheries research

1. TERM OF THIS SOW

(clause 4.2 Error! Reference source not found. **of Schedule 2)**

Commencement Date: 10 September 2026

End Date: 30 June 2028

2. CONTACT DETAILS

(refer clause 23 of Schedule 2)

The initial contact persons for each Party for the purposes of this SOW are as set out below. If a Party's contact persons or their details change, that Party must notify the other Party in accordance with clause 23 of Schedule 2.

	Contract Manager	Technical Liaison
MPI	Fetuao Nokise Manager, Fisheries Contracts Fetuao.Nokise2@mpi.govt.nz 04 894 0052	Jean Davis Senior Scientist, Aquatic Environment Jean.Davis@mpi.govt.nz 03 266 1989
Contractor	Adam Smith Director Sea Through Science Adam@seathrough.science 021 0383941	Adam Smith Director Sea Through Science Adam@seathrough.science 021 0383941

3. SERVICES

3.1 Service Description

The Services described below are within the permitted scope for Panel Services, as set out in the Panel Services Agreement,

The Contractor will perform the Services and provide the Deliverables set out in the table below. Each Service and Deliverable is referred to in this SOW by the name given to it in the table below.

Service (Name and description)	Deliverable (Name and description)
Overall Objective	To analyse and report on trends in fish abundance and biodiversity from baited remote underwater videos (BRUVs) collected in Fiordland and complete further field work to test the power of the method for detecting trends in blue cod abundance.
Objective 1	Annotate BRUVs collected in 2024 and 2025, recording data on fish abundance and length information, with a focus on blue cod and Hapuku.
Objective 2	Design and implement a field experiment to test the viability of BRUVs to detect changes in blue cod abundance.
Objective 3	Analyse all BRUV data since 2024 and the results of the field work done in Objective 2 to report on the power of the method to detect change in fish populations.
Objective 4	Economic Benefits to New Zealand

3.2 Notice if Research Objectives at Risk

If the Contractor becomes aware of any circumstances that materially reduce the likelihood of the Research Objectives being achieved, it must as soon as possible notify the MPI Technical Liaison in writing of those circumstances.

3.3 Dependencies

Completion of the following Services or Deliverables by the Contractor is dependent on the persons below performing their associated dependencies below. **Note:** The failure of a dependency is treated as an "Extraordinary Event" affecting the Contractor, to which the provisions of clause 25 (Extraordinary Events) of Schedule 2 apply.

Service/ Deliverable	Dependency	Person Responsible
All specific objectives	MPI data is made available for the provision of Services	MPI Technical Liaison

3.4 Key Personnel

The Contractor's Key Personnel are:

Position Title	Name	Estimated Time	Role
Principal Investigator:	Adam Smith	394 hours	Principal Scientist, reviewer and approver of outputs for release.
Field Assistant/Technican	Alex Burton	271 hours	Field work and annotation of stereo-BRUV footage
Technician	Heidi Schlumpf	120 hours	Annotation of stereo-BRUV footage

3.5 Research Activities/Services

The Contractor will perform the activities set out in the table below. The Contractor will complete the Services in accordance with the summary below.

Objective One: Annotate BRUVs collected in 2024 and 2025, recording data on fish abundance and length information, with a focus on blue cod and Hapuku.

All annotations will be done in *EventMeasure* (SeaGIS Ltd), the industry standard software for extracting counts and calibrated lengths from stereo-video, following the approaches established under SEA2023-15 for the 2024 deployments, and general guidelines by Langlois et al. (2020). Each annotation period begins five minutes after the unit settles on the seafloor, to allow disturbed sediment to clear, and runs for 60 minutes. We will record two indices of relative abundance: *MaxN*, the maximum number of individuals of a species in a single stereo frame, which avoids double-counting and remains comparable across species and with the earlier series; and *MeanCount*, the average count across seven systematic frames at 0, 10, 20, 30, 40, 50 and 60 minutes, which may scale more nearly linearly with abundance (Campbell et al. 2015; Schobernd et al. 2014) for common species such as blue cod (*Parapercis colias*). Lengths will be measured for blue cod, hāpuku (*Polyprion oxygeneios*), tarakihi (*Nemadactylus macropterus*), school shark (*Galeorhinus galeus*) and broadnose sevengill shark (*Notorynchus cepedianus*), from the MaxN frame and from each of the seven systematic frames; lengths from the systematic frames may reduce biases that arise when MaxN falls at a moment dominated by a single size class.

Every annotation will be reviewed by the Principal Scientist before the data are accepted, covering species identity, counts, and the plausibility of each length measurement against its stereo geometry; taxa that cannot be confidently identified will be captured as screenshots and reported as unresolved rather than assigned. Where AI-assisted detection and measurement tools are mature enough to improve consistency or throughput (Marrable et al. 2023), we may use them in support of annotation, with all output verified manually before acceptance. Annotated data will be delivered as EventMeasure measurement files together with a database export and deployment metadata linked by OpCode, as detailed in our SEA2023-15 report and data submission.

The annotations of 2024 deployments completed under SEA2023-15 recorded abundance for the focal species only, with other taxa noted as present or absent. In this project we will record MaxN for every species observed. Objective 3 requires trends in biodiversity across sites and years for all species observed in the videos, and assemblage analyses need counts rather than species lists.

Objective 3 develops a framework for analysing BRUV data to examine trends in abundance, and the annotation protocol is designed to supply the data required under that framework. Part of this process will be identifying the covariates that influence what a BRUV records (e.g., depth, habitat, current, and water clarity) and finding practical ways to measure these from the footage. These covariates will be used to develop a standardised index of abundance that allows for conditions that vary between years, much as catch and effort data are standardised for a fishery. Some of these measures are now in routine use — for example, we now classify water clarity for every deployment on a three-level scale (1–3 m, 3–7 m, 7+ m), rather than counting fish only within a fixed distance of the cameras — and others will be developed under this project. These same covariates will need to be collected for all deployments to be analysed as part of Objective 3.

Under this objective, we will annotate the 102 deployments from the February 2025 campaign (FLD25) that remain outstanding in Tamatea/Dusky Sound, completing the annotation of that campaign in full. In addition, we will update the 2024 series with MaxN counts and all the necessary covariates. This will yield approximately 140 annotated successful deployments in Tamatea/Dusky Sound in each of 2024 and 2025, roughly 90% of which were the same stations surveyed in both years. The

remaining 125 unannotated deployments from 2021 and 2025 are offered as separately priced additional scope in the Pricing Schedule; all footage is archived and is available for annotation at any time.

Table 1. Annotation status of the Fiordland stereo-BRUV deployments, August 2026, with annotations to be completed under this project in red. One deployment in Breaksea Sound (2025) and three in Dusky Sound (2024) failed and are excluded. DOC funded annotation of some of the FLD25 deployments. For FLD24, all 136 usable deployments have been annotated; 42 are already at the current protocol and 94 require updating, which means bringing them to it (MaxN for all species; complete covariates) rather than annotating from scratch. "Other" in the case of FLD21 includes deployments in Doubtful Sound / Patea, Taiari / Chalky Inlet and Rakituma / Preservation Inlet.

	FLD25	FLD25	FLD24	FLD24	FLD21	FLD21
Area	Annotated	To do	Annotated	To update	Annotated	To do
Dusky	42	102	42	94	0	24
Breaksea	0	35	–	–	0	12
Wet Jacket	24	0	–	–	0	12
Other	0	0	–	–	0	42
TOTAL	66	137	42	94	0	90

Objective Two: Design and implement a field experiment to test the viability of BRUVs to detect changes in blue cod abundance.

Under this objective, we will run a paired comparison of pots and BRUVs in which each station is sampled once by each gear, with the order of gears counterbalanced across stations. Three pots and three BRUVs will be deployed alternately (BRUV-pot-BRUV-pot-BRUV-pot) across sets of six stations, spaced at least 400 m apart; on a later day, each set of stations will be revisited and sampled in the same order with the other gear (pot-BRUV-pot-BRUV-pot-BRUV). The design means that each station serves as its own "control", yielding an experimental design better-suited to our aims than simply deploying each gear at a different set of sites. MaxN for blue cod across the annotated 2025 deployments spans 0 to 31, and a between-groups comparison would have to estimate the gear effect against the among-station variation. The replication unit is the station rather than the six-pot set used in the historical potting surveys; sets comparable with the potting series can be reconstructed analytically afterwards, whereas the paired structure cannot be recovered if it is not built into the field design.

There is a potential carryover effect from the first deployment to the second (alternate gear) deployment at each station. Blue cod are site-attached, so fish caught and released on the first visit remain to be caught on the second; trap-shyness would depress the pot catch where pots came second, while a bait plume left by a BRUV could aggregate fish where the pot came second. However, we will guard against this issue by (1) leaving at least one day between deployments and, more importantly, (2) counterbalancing the methods – some stations will have BRUV first and pot second, and other stations vice versa. This means that any carryover effect is not confounded with the gear effect.

Stations will be revisited in the same order, so that each is sampled at approximately the same time of day under both gears; the potting manual advises against standardising for time of day when constructing an abundance index, but for a within-station gear comparison, matching is strictly better than randomising.

The study will take place in outer Dusky Sound, where we have sound knowledge of the habitats and fish assemblages from past BRUV surveys. We will define three strata, with the aim of getting a range of abundance levels and size classes. Among the annotated 2025 deployments, blue cod were consistently most abundant shallower than 40 m and scarce below about 70 m; we will therefore restrict stations to depths under 40 m. If conditions permit, one stratum will be on the outer, exposed coast where densities of blue cod are generally higher.

We have costed the fieldwork for 36 paired stations, based on two sets of six deployments per day across six deployment days within a seven-day vessel period in July-September 2027 (allowing one day for transit, maintenance, and other contingencies). If time and success allow, we may be able to achieve three sets of deployments per day and raise the sample size. Fieldwork will be conducted from the Department of Conservation vessel Southern Winds within the July to September 2027 availability window advised by the Department. All our previous BRUV campaigns in Fiordland used this vessel, and it is well-equipped for both BRUVing and potting.

Pot deployment will follow the per-pot protocol of the Fisheries New Zealand blue cod potting manual as far as possible while keeping to the objectives of the study. In our study, the pots will yield counts and lengths only, with no otoliths and no dissection-based sex determination.

Soak times will be standardised to 65 minutes from when the gear hits the seafloor. Lifting pots on time will be the priority because the annotation period of BRUVs is standardised precisely regardless of extended soak times.

We will aim to install calibrated stereo cameras on the pots, which, if successful, will provide supplementary information about the size structure and behaviour of fish around the pots, particularly entry and exit. This doubles the potential annotation workload in the context of a full project, so annotation of a subset of the pot-attached footage is offered as additional scope in the Pricing Schedule. The success of Objective 2 is not contingent on pot-video data.

Several administrative tasks will be completed prior to the commencement of field operations. The study design will be presented for approval to the Fiordland Marine Guardians and the Biodiversity Research Advisory Group, as required under the reporting milestones. The work will require a special permit issued under section 97 of the Fisheries Act, and because potting involves handling live animals, approval from a university ethics committee will be sought. A health and safety management plan specific to the field programme will be submitted for approval at least one month before mobilisation.

Objective Three: Analyse all BRUV data since 2024 and the results of the field work done in Objective 2 to report on the power of the method to detect change in fish populations.

Baited underwater video has its origins in New Zealand, where it was developed as a non-destructive alternative to visual census and angling for surveying carnivorous reef fish (Willis & Babcock 2000; Willis et al. 2000). Since then, its use has been somewhat limited, particularly in fisheries contexts. Underwater video is well suited to environments such as Fiordland, where the established survey method is destructive, much of the fished habitat is deep or exposed, and diver-based methods are costly and impractical. Yet, whether it can take the place of potting has not been demonstrated quantitatively. Here, we will analyse the Fiordland stereo-BRUV data, together with the paired deployments made under Objective 2, to test whether stereo-BRUVs can serve in place of pots for monitoring the species of interest in Fiordland. Specifically, we will examine whether a BRUV-based index can estimate the relative abundance of key species with precision comparable to that achieved by the potting survey, and whether the size structure it records is consistent with that obtained by potting at the same stations.

The difficulty in constructing such an index is that the appearance of fish in any particular video is influenced by proximate, context-specific covariates such as visibility, occlusion, camera angle, habitat structure and water movement (Bacheler et al. 2014), which can introduce noise on top of genuine ecological patterns, or even falsely suggest change where none has occurred. Such factors exist for any sampling method, but one advantage of video over traps (Bacheler & Shertzer 2020) is that many can be known: recorded from the footage for each deployment and included in the analysis.

The analysis will use the annotated 2024 and 2025 Tamatea/Dusky Sound deployments described under Objective 1, the 2021 deployments being additional scope, together with the paired-stations experiment from Objective 2. From these data we will report on the abundance, size structure and variability of blue cod and the other key fished species with respect to depth, habitat, protection status and year, and on broader trends across the wider assemblage, which recording MaxN for every species now makes possible. Reserve contrasts are observational and confounded with depth and habitat, so they are adjusted comparisons rather than experiments.

We will develop a standardisation framework for stereo-BRUV data in Fiordland, analogous to the established methods used to standardise catch-per-unit-effort series in fisheries science (Maunder & Punt 2004), and applied to video-survey counts elsewhere (Bacheler et al. 2023). Analyses will be conducted in R, using count-based generalised linear mixed models, likely negative binomial and allowing for zero inflation where required (Smith et al. 2012, 2014), fitted with Bayesian approaches via brms and Stan. An index derived from BRUVs is a relative measure rather than an absolute one, and as such, comparisons drawn from it hold only insofar as the biases inherent to the method are stable across the comparison being made. We will therefore examine both indices recorded under Objective 1 (MaxN and MeanCount) rather than commit in advance to one, since which better serves the monitoring task cannot be settled by appeal to true abundance: every measure available to us is itself an index. We can, however, compare the precision of the indices. The potting survey of Dusky Sound provides the benchmark with a coefficient of variation of 11.9% on the stratified mean from 28 fixed sites and 8.6% from 34 random sites (Beentjes & Page 2016).

Further, the paired stations from Objective 2 allow us to compare BRUV and pot indices more directly. We will examine whether a video index and pot catch differ by a factor that is stable across depth, water clarity and density, which is what substitution for the potting series would require, and we will compare the length compositions obtained by the two gears at the same stations. The catchability of pots is itself known to vary with temperature and substrate (Bacheler & Shertzer 2020), so stability cannot be assumed for either gear.

We will simulate from the fitted models to estimate the number of stations required to achieve benchmark precision for abundance, and the change detectable at a given number of stations, reported across a range of station numbers rather than as a single figure. The same simulations address the survey-design refinement this objective calls for: the number of spatial strata, and the number of stations within each. Code developed under the project will be supplied in standalone form as part of the data submission, such that a third party can reproduce the supplied outputs from the supplied inputs.

3.6 Reports

The Contractor will report to MPI as follows, and as otherwise reasonably required by MPI, in any format reasonably required by MPI and at no cost to MPI:

Type of Report	Report To	Due Date
Work in progress (WIP) Report	Contracts Monitoring (business.operations@mpi.govt.nz)	3rd Business day of each month

MPI may request documentary evidence from the Contractor in relation to any item reported against.

3.7 Milestones

Each Party will meet its Milestones set out in the table below. The Price set out in the table below will be invoiced in accordance with clause 7 of Schedule 2 (Price).

MS#	Obj#	RR#	Milestone Description	Due Date	Fixed Price (plus GST)
1	1-3	1	To hold a project scoping meeting to discuss project goals and decision points with MPI	30 September 2026	$8,000
2	1-3	2	To hold an internal meeting with MPI, and DOC to discuss the methods for Objective 2	31 October 2026	$10,000
3	1-3	3	To present project methods to the Fiordland Marine Guardians (FMG), the Fisheries New Zealand Biodiversity Research Advisory Group (BRAG) and other relevant working groups	30 November 2026	$20,000
4	2		Submit health and safety management plan for approval 1 month before activity commences.	31 May 2027	$0
5	1	4	Submit a short summary report on the annotation of historic data (Objective 1) and deliver the annotated dataset	30 June 2027	$89,000
6	1-3	5	To submit to the MPI Contracts Monitoring and Administration Team, Fisheries New Zealand Science Officer, and Fisheries New Zealand project manager a draft Aquatic Environment and Biodiversity Report (AEBR) (as per Research Reporting Form 6)	15 December 2027	$34,000

			summarising all project findings and present final results to the BRAG working group with a Form 9: Research Report Submission Coversheet.		
7	1-3	6	To submit a final AEBR (as per Research Reporting Form 6) to the MPI Contracts Monitoring and Administration Team, Fisheries New Zealand Science Officer, and Fisheries New Zealand project manager with a Form 9: Research Report Submission Coversheet.	31 March 2028	$5,000
8	1-3		To submit any data generated, collected or modified during the course of this project to Fisheries New Zealand Data Management (fnzdatasubmission@mpi.govt.nz)	31 March 2028	$8,800
				Total Price	**$174,800**

4. PRICE

(clause 7, Schedule 2)

4.1 Fixed Price

The Contractor may invoice MPI for the Fixed Prices in clause 3.7 (Milestones) up to a total of $174,800.00 (plus GST if any). Each invoice may be issued when the invoicing criteria related to that invoice has occurred to MPI's reasonable satisfaction.

5. INTELLECTUAL PROPERTY

(clause 16 Schedule 2)

For the purposes of clause 16 of Schedule 2, the Parties agree that "Option A" and the "Compulsory Clauses" shall apply, but that "Option B" is deemed deleted from this SOW.

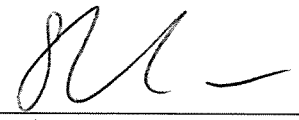
6. VARIATIONS TO PANEL AGREEMENT

(clauses 3 and 13, Schedule 2)

None.

SIGNATURES

SIGNED for and on behalf of **MPI** by the person named below, being a person duly authorised to enter obligations on behalf of MPI:



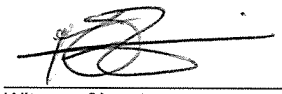
Signature

Name: Simon Lawrence

Title: Director

Date: 17/9/26

in the presence of:



Witness Signature

Name: Fetuao Nokise

Occupation: Manager, Fisheries Contracts

Address: 34-38 Bowen Street
Charles Ferguson
Building

SIGNED for and on behalf of the **Contractor** by the person named below, being a person duly authorised to enter obligations on behalf of the Contractor:



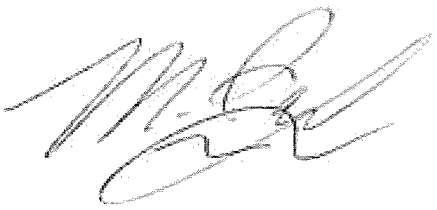
Signature

Name: Adam Smith

Title: Director, Sea Through Science Ltd.

Date: 16 September 26

in the presence of:



Witness Signature

Name: Matthew Pawley

Occupation: Associate Professor, Auckland University of Technology

Address: 30 Marua Road, Auckland

DATA MANAGEMENT PLAN

ALL DATABASE DOCUMENTATION IS AVAILABLE ON THE MINISTRY FOR PRIMARY INDUSTRIES MARLIN WEBSITE AT:

https://marlin.niwa.co.nz/databases

MPI shall own all the data collected or created during the course of the project in whatever form or format. All data should be submitted to the Ministry's Research Data Manager ahead of the contracted data milestone due date(s) or no later than the final milestone due date. This Schedule has been developed to determine which data types will be expected by the Ministry's Research Data Management (RDM) team at the time of submission.

The type of data that might be produced or used as part of a project may include:

1. electronic data as described in Section 1 (associated with a FNZ database),
2. physical data as described in Section 2 (specimens, paper), or
3. other data as described in Section 3 (not covered by sections 1 and 2).

Explicit metadata must accompany all datasets at time of data submission. An exception to this is the production of new data for FNZ databases (Section 1), which is constrained by the database structure as described in the publicly available documentation.

NOTE:

- The data submission should serve as a stand-alone project specific resource, including all inputs and outputs.
- If using **existing data** described in Section 1 (electronic) or Section 3 (other data) in analyses or to produce an output, the original data extract, along with extract references (e.g. replog number), should be included in the data submission.
- Any **computer code** developed or utilised under the project, including code to extract data from databases, code to analyse data, code to groom data, and code to produce any outputs, must be included in the data submission and described in supplied metadata.
  The code should enable a third party to reproduce the supplied outputs from the supplied inputs. Code must be supplied as a standalone format, i.e links to code stored in third party code repositories (e.g GitHub, GitLab, BitBucket)will not be accepted.
- Matauranga Māori data refers to information or knowledge in a digital or digitisable form that is about or from Māori peoples and our environments, regardless of who controls it. Maori data needs to be stored and transferred in such a way that it enables and reinforces the capacity of Māori to exercise kaitiakitanga over Māori data. Māori should decide which Māori data sets should be controlled (tapu) or open (noa) access.Tikanga, kawa (protocols) and mātauranga Māori (knowledge) should underpin the protection, access and use of Māori data.

  Examples include Interviewing to gather knowledge, documenting knowledge as physical media – surveys, maps, images etc

SCHEDULE OF DATA

TO BE PROVIDED TO THE MPI RESEARCH DATA MANAGER

Contact Name *(for data queries):*	Adam Smith	**Contract No:**	407609
Contact Email *(for data queries):*	adam@seathrough.science	**Project Code:**	INV2026-01

The following questions will guide you in determining which boxes to tick. Tick all boxes that apply to the work proposed in your statement of work and **please provide extra information as required.** For each box ticked, you will be expected to submit the corresponding data in the correct format.

1. Electronic data (database) *For this section, you will either be collecting new data to upload to an existing database and/or extracting and using data from a database. If collecting new data, the data must be submitted in the exact structure described in the associated database documentation for upload to the database (click hyperlinks to get to database documentation). The format should be in comma separated values with one file per database table. If using, analysing or grooming existing data - the source, workings, and output data for that process must be submitted .*	**Associated database**	**Collecting or producing new data**	**Using, analysing or grooming existing data**	**Mātauranga Māori data or information collected**	**For FNZ admin use**
a. Using data from commercial catch effort returns as stored on the Enterprise Data Warehouse.	EDW		☐	☐	GRM01
b. Using data collected on NZ vessels fishing outside the NZ EEZ.	regional_ce		☐	☐	GRM01
c. Using data collected under the MPI Observer programme	cod		☐	☐	GRM02 GRM03 GRM05
d. Acoustic survey data on distribution and abundance of marine species.	acoustic	☐	☐	☐	DBS01
e. Specimen ageing data (e.g. data derived from otolith collection, reading)	age	☐	☐	☐	DBS03
f. Data about the biological characteristics of individual marine organisms using transect surveys of **beaches**	beach	☐	☐	☐	DBS04
g. Marine Biodiversity and Biosecurity (distribution, location or abundance of marine organisms or physical/chemical properties of the sea floor, substrate or water column).	biods	☒	☒	☐	DBS05 DBS06
h. Conductivity, temperature and depth of the water column using a CTD device.	ctd	☐	☐	☐	DBS08
i. Dive surveys to determine biological characteristics of individual marine organisms.	dive	☐	☐	☐	DBS09
j. Partially groomed commercial catch and effort data from deepwater fisheries (orange roughy and oreo) used for stock assessments.	dw_cdb	☐	☐	☐	GRM01
k. Data about the biological characteristics of individual species by **onshore sampling** (usually in a processing facility or shed (excluding rock lobsters).	market	☐	☐	☐	DBS15
l. Data about the biological characteristics of individual **oysters** using dredge surveys.	oyster	☐	☐	☐	DBS19

m. Data about the biological characteristics of individual marine organisms using surveys of **recreational fishing** (e.g. fishing diaries, ramp surveys and shellfish harvesting surveys).	rec_data	☐	☐	☐	DBS22
n. Data about the biological characteristics of individual post puerulus **rock lobsters** on commercial vessels or by market sampling.	rlcs_a	☐	☐	☐	DBS24
o. Data about the biological characteristics of individual post puerulus **rock lobsters** collected by commercial fishers in logbooks.	rlcs_b	☐	☐	☐	DBS24
p. Data about the biological characteristics of individual rock lobsters by **phyllosoma or puerulus sampling.**	rocklob	☐	☐	☐	DBS25
q. Data about the biological characteristics of individual **scallops** using dredge surveys.	scallop	☐	☐	☐	DBS26
r. Data on seamounts	seamount	☐	☐	☐	DBS27
s. Data about the biological characteristics of individual marine organisms using **tagging** programmes.	tag	☐	☐	☐	DBS28
t. Data about the biological characteristics of individual marine organisms using **trawl** surveys.	trawl	☐	☐	☐	DBS29
u. Tuna catch effort data derived from commercial returns and covers New Zealand, Australian, and Japanese tuna fisheries.	tuna	☐	☐	☐	DBS30
v. Protected species capture data (no documentation available). (could be the psc database created under contract with Proteus and stored at NIWA (currently under negotiation)	protected species captures	☐	☐	☐	PSC
w. Merging of electronic datasets extracted from databases described in this section. For example, combining commercial catch effort and observer data. *Specify the databases*	Please specify: _____	☐		☐	GEN05

2. Physical data (specimens, paper) *Explicit metadata is required with submission of these datatypes. Corresponding database(s) in Section 1. above must be selected as appropriate.*	**Specify type**	**Collecting or producing new data**	**Using existing physical data**	**Mātauranga Māori data or information collected**	**For FNZ admin use**
a. Specimens, ageing (types: otolith, scale, verterbrae, teeth, spine). Note: Specimens must be sent/returned to NIWA (e,g, the 'otolith store' for dry samples) with data submission.	Otoliths	☐	☐	☐	SPC01
b. Specimens, non-ageing (types: stomach contents, gonad, genetics, etc).		☐	☐	☐	SPC02

Expected fate of samples: _____					
c. Whole organisms (types: fish, invertebrate, tissue, etc). Expected fate of samples: _____		☐	☐	☐	SPC03
d. Substrate (types: mud, rocks, etc) Expected fate of samples: _____		☐	☐	☐	IMG02
e. Physical paper forms (types: survey, diary, etc). Note: Forms must be sent/returned to NIWA physical data store with data submission.	Sampling forms	☐	☐	☐	RAW01

3. Other data not described above *Electronic data submissions should only be included here if not appropriate for upload to a database described in section 1. Explicit metadata required with submission of these data types.*	**Specify type**	**Collecting or producing new data**	**Using existing data** RAW02	**Mātauranga Māori data or information collected**	**For FNZ admin use**
a. Audio / visual data (types: audio, images, video, graphs).		☒	☒	☐	IMG01
b. Fishing equipment data (types: trawl, longline, recreational, etc)		☐	☐	☐	XTR02
c. Geospatial/Mapped data (types: shapefiles, geodatabases, etc). Note: maps within project reporting should have accompanying geospatial files. Latitude/longitude fields within spreadsheets do not qualify as geospatial data in this context.		☐	☐	☐	IMG03
d. Electronic genetic data including any data from/for genetics db (types: raw sequencing data, etc).		☐	☐	☐	DBS11
e. Online data publication – uploading newly produced data or downloading existing data (types: OBIS, GBIF, GenBank, etc). Note: Limited to publication of project data on reputable data portals, excludes MPI publishing the report and/or metadata.		☐	☐	☐	XTR05
f. Any other grooming, manipulation or modelling of electronic data not already described above.		☒		☐	GEN01
g. Any other data types not described above. May include intermediate-stage, working or other processing files used to derive outputs. *Specify*		☐	☐	☐	DBS02 DBS10 DBS12

associated databases if applicable e.g. less common FNZ research datasets not listed above can be seen here. Or details of use or merging with external datasets, etc: _____ _____ _____ _____					DBS13 DBS14 DBS16 DBS18 DBS20 DBS21 DBS23 GEN01 GEN02 GEN03 GEN04 OBS17 RAW02 XTR01 XTR04

ATTACHMENT 2: CURRICULUM VITAE

CURRICULUM VITAE	
Full Name:	ADAM NICHOLAS HOWARD SMITH
Present Position:	Founder and Director; Statistical Consultant
Present Employer:	SEA THROUGH SCIENCE LIMITED
Present Work Address:	39 Amorino Drive, Red Beach, Auckland 0932, New Zealand
Academic Qualifications:	2016 PhD in Statistics, Massey University. Thesis: *Bayesian Modelling of Direct and Indirect Effects of Marine Reserves on Fishes* 2006 MSc (Hons) in Statistics, University of Auckland. Thesis: *Evaluation of the New Zealand Marine Environment Classification for Shallow Coastal Rocky Reef Fish Communities* 2003 PGDipSci (Distinction) in Ecology, University of Canterbury 1998 BSc in Zoology, University of Canterbury
Years as a Practising Researcher:	24
Professional Positions Held:	2023 – present · Founder, Director and Statistical Consultant, Sea Through Science Limited, Auckland 2017 – present · Course Instructor and Statistical Consultant, PRIMER-e, Auckland 2013 – 2025 · Lecturer, then Senior Lecturer in Statistics, Massey University, Auckland 2006 – 2008 · Fisheries Scientist, National Institute of Water and Atmospheric Research, Wellington 2002 – 2006 · Ecological Statistician, Department of Conservation, Christchurch
Present Research/Professional Speciality:	Statistical modelling for fisheries and marine ecological monitoring. Bayesian hierarchical modelling of sparse, over-dispersed and zero-inflated count data of the kind produced by fishery-independent surveys; multivariate analysis of species assemblage structure; and the design, analysis and reporting of stereo baited remote underwater video (stereo-BRUV) surveys. Reproducible analysis and reporting workflows in R, Quarto and Stan.
Experience Relevant to Proposal:	Led multiple stereo-BRUV campaigns, including those in Fiordland (FLD21, FLD24, FLD25), over 10 years, covering survey design, fieldwork from the Department of Conservation vessel Southern Winds, annotation oversight and reporting. Principal Investigator on SEA2023-15 for Fisheries New Zealand, annotating the 2024 Dusky Sound deployments and reporting to the Biodiversity Research Advisory Group. Author of the July 2025 status report on the Fiordland stereo-BRUV programme for the Department of Conservation. Co-author of the international field and video-annotation guide for baited remote underwater stereo-video surveys (Langlois et al. 2020) and contributor to Global FinPrint (MacNeill et al 2020). Doctoral research applied Bayesian zero-inflated models to baited underwater video counts of fishes in marine reserves — the same class of model proposed under Objective 3. Stock assessment modelling for pāua and rock lobster at NIWA. Ten years of stereo-BRUV work across five countries; extensive vessel-based marine fieldwork in New Zealand, Oman, New Caledonia, Fiji and the Galápagos Islands.
Relevant Publications:	Smith, A.N.H.; Anderson, M.J.; Millar, R.B. (2012). Incorporating the intraspecific occupancy–abundance relationship into zero-inflated models. *Ecology* 93: 2526–2532. Smith, A.N.H.; Anderson, M.J.; Millar, R.B.; Willis, T.J. (2014). Effects of marine reserves in the context of spatial and temporal variation: an analysis using Bayesian zero-inflated mixed models. *Marine Ecology Progress Series* 499: 203–216.

	Smith, A.N.H.; Anderson, M.J. (2016). Marine reserves indirectly affect fine-scale habitat associations, but not overall densities, of small benthic fishes. *Ecology and Evolution* 6: 6648–6661. Smith, A.N.H.; Duffy, C.A.J.; Leathwick, J.R. (2013). Predicting distribution and relative abundance of fishes on shallow subtidal reefs. *Science for Conservation* 323. Department of Conservation, Wellington. Langlois, T.; Goetze, J.; Bond, T.; Monk, J.; … Smith, A.N.H.; … Harvey, E.S. (2020). A field and video-annotation guide for baited remote underwater stereo-video surveys of demersal fish assemblages. *Methods in Ecology and Evolution* 11: 1401–1409. MacNeil, M.A.; Chapman, D.D.; Heupel, M.; Simpfendorfer, C.A.; Heithaus, M.; Meekan, M.; … Smith, A.N.H.; et al. (2020). Global status and conservation potential of reef sharks. *Nature* 583: 801–806. Full list: https://www.seathrough.science/publications/

CURRICULUM VITAE	
Full Name:	ALEX J. C. BURTON
Present Position:	Laboratory Decommissioning Technician (Chemistry), Massey University; annotation and field subcontractor to SEA THROUGH SCIENCE LIMITED
Present Employer:	Massey University, Auckland
Present Work Address:	Massey University, Albany Campus, Auckland, New Zealand
Academic Qualifications:	2026 PhD, Massey University — multidisciplinary thesis in marine ecology, chemistry and statistics on the life-history and trace element dynamics of school sharks (*Galeorhinus galeus*) Postgraduate Diploma in Science and Technology, Massey University — advanced chemistry, statistics and wildlife management Bachelor of Science (Massey Scholar), Massey University — major in Marine Ecology, minors in Chemistry and Statistics
Years as a Practising Researcher:	6
Professional Positions Held:	2025 – present · Laboratory Decommissioning Technician (Chemistry), Massey University, Auckland 2025 · Field assistant, stereo-BRUV deployments, contracting to Sea Through Science Limited (February and April) 2024 – 2025 · Annotator, stereo baited remote underwater video, contracting to Sea Through Science Limited (March–September 2024; April–July 2025)
Present Research/Professional Speciality:	Marine ecology and fisheries science with a quantitative and analytical-chemistry background. Life-history and movement of elasmobranchs; statistical modelling in R and Stan, including length-conversion models and meta-analysis of life-history traits; elemental analysis of biological tissues; and the deployment and annotation of stereo baited remote underwater video.
Experience Relevant to Proposal:	Annotated stereo-BRUV footage for Sea Through Science under SEA2023-15 (the 2024 Tamatea/Dusky Sound deployments) and under the DOC-funded annotation of the 2025 campaign, counting and measuring fish in EventMeasure and AFID and assessing habitat from the footage. Co-developed the EventMeasure annotation procedure now used across the Fiordland programme, so no training in the protocol is required for this project. Field assistant on the February 2025 Fiordland campaign aboard the Department of Conservation vessel *Southern Winds*, assembling and checking units, preparing bait, rigging ropes and buoys, and deploying and retrieving BRUVs. Further stereo-BRUV deployment experience off the Otago coast (RV *Papanui*) and in the outer Hauraki Gulf (MV *Seasprite II*) in 2025. Holds Boatmaster and Maritime VHF certification. Presented life-history work to the Fisheries New Zealand Inshore Working Group in 2025, and has held two Ministry for Primary Industries scholarships.
Relevant Publications:	Burton, A.J.C.; Braccini, J.M.; Smith, A.N.H. (in press). Navigating morphometric minefields: modelling heteroscedasticity in length-conversion models. Journal of Fish Biology. Burton, A.J.C.; Duffy, C.A.J.; Tindale, S.C.; Griffiths, M.A.; Walker, T.I.; Smith, A.N.H. (in preparation). Broad dispersal among key habitats in a lifeboat population of school sharks (*Galeorhinus galeus*).

	Burton, A.J.C.; Harrison, J.A.; Walker, T.I.; Smith, A.N.H.; Thelen, M.A. (in preparation). Is it my burden to bear: the maternal transfer potential of school sharks (*Galeorhinus galeus*). Burton, A.J.C. (2020). Overview of the monitoring and research in the Long Bay – Ōkura Marine Reserve: a report of studies conducted up till December 2019. Department of Conservation internal report.

CURRICULUM VITAE	
Full Name:	HEIDI ANN SCHLUMPF
Present Position:	Annotator and Data Administrator (shareholder)
Present Employer:	SEA THROUGH SCIENCE LIMITED
Present Work Address:	39 Amorino Drive, Red Beach, Auckland 0932, New Zealand
Academic Qualifications:	In progress — Bachelor of Health Science (Nursing), Auckland University of Technology 1999 — BSc (Hons), First Class, in Zoology, University of Canterbury 1998 — BSc, double major in Zoology and Psychology, University of Canterbury
Years as a Practising Researcher:	24 years of professional scientific and technical practice
Professional Positions Held:	2025 – present — Annotator and Data Administrator, SEA THROUGH SCIENCE LIMITED, Auckland 2020 – 2025 — Project Manager, Digitisation, New Zealand Maritime Museum Hui Te Ananui a Tangaroa, Auckland 2016 – 2020 — Senior Collection Technician, Auckland War Memorial Museum Tāmaki Paenga Hira 2011 – 2016 — Palaeontology Project Technician, Auckland War Memorial Museum Tāmaki Paenga Hira 2005 – 2007 — Palaeontology Curator, GNS Science Te Pū Ao, Wellington 2002 – 2005 — Collections Technician, Canterbury Museum, Christchurch
Present Research/Professional Speciality:	Annotation of stereo baited remote underwater video, and the management of scientific data and records. Specialist in developing and enforcing data standards across teams — cataloguing methodology, version control, quality control, and collections database administration — applied here to the consistency and completeness of annotation datasets.
Experience Relevant to Proposal:	Annotates stereo-BRUV footage for SEA THROUGH SCIENCE in EventMeasure, including deployments from the Fiordland campaigns under contract to the Department of Conservation, surveys for Northland Regional Council, and Hauraki Gulf surveys for the Department of Conservation. Already trained in the annotation protocol used across the Fiordland programme, so requires no lead-in for this project. Twenty years of managing large scientific datasets and the standards that keep them consistent: led the Natural Sciences Collections Readiness Programme at Auckland Museum, coordinating cataloguing, imaging and specialist taxonomy workflows and supervising a team of up to ten technicians across five collection areas; led two digitisation programmes at the New Zealand Maritime Museum funded by the New Zealand Lotteries Commission and Tātaki Auckland Unlimited, covering end-to-end coordination, records management, quality control, staff training and supervision, and budget administration. Developed and documented methodologies to ensure consistency, and maintained document version control across internal and external teams. Administered the Vernon collections management database at three institutions and contributed to database development and customisation. Elected Health and Safety Representative at Canterbury Museum for two years.
Relevant Publications:	Trnski, T.; Schlumpf, H. A. (eds) (2015). *Kermadec Biodiscovery Expedition 2011*. Bulletin of the Auckland Museum 20. Auckland War Memorial Museum, Auckland. Scofield, R.P.; Worthy, T.H.; Schlumpf, H.A. (2003). What birds were New Zealand's first people eating? Wairau Bar's avian remains re-examined. *Records of the Canterbury Museum* 17: 17–35.